

Animal Intrusion Detection Using Yolo V8

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Abstract—This project presents an inventive Animal Intrusion Detection System using Yolo v8 object detection algorithm. Addressing the problem of wildlife intrusions in agriculture as its main goal, the presented work focuses on precise animal identification and the activation of appropriate warning systems. The technology effectively identifies animals in agricultural fields, protecting crops as well as guaranteeing the welfare of wildlife. An intelligent monitoring system that can detect the image of an animal entering, sending alert messages to the farmers or landlords is proposed. It may lessen conflicts between people and wildlife, safeguard crops, and maintain biodiversity, all of which would be advantageous to both wildlife conservation and agriculture. The presented work uses YOLO v8, the most recent instantaneous object detector in the YOLO family, because it performs exceptionally well in terms of accuracy and speed. YOLOv8 offers new features and enhancements, building on the strengths of earlier YOLO versions. This makes it perfect for object detection in a range of applications.

Keywords—animal intrusion detection; yolo v8; wildlife protection, agriculture; crop damage prevention; wildlife safety; human-wildlife conflicts; biodiversity conservation; precise animal recognition; agriculture protection;

I. INTRODUCTION

India has historically depended heavily on agriculture, with the International Labor Organization estimating that 60% of the country's population works in the agricultural sector. Nonetheless, the nation's farmers deal with a number of difficulties, the primary one being confrontations between humans and animals that endanger human life and cause large resource losses. In order to avoid animals and other undesirable incursions, it is imperative that these conflict zones be continuously monitored in order to find a solution. Thanks to the nation's long history of conservation, around one-fifth of India's territory has been set aside as national parks, game reserves, animal sanctuaries, or protected areas. India has gained recognition for its efforts to involve nearby people in the administration of these protected areas. But there have also been disputes between people and animals as a result of this behaviour. Numerous significant manifestations of these conflicts have been documented via studies conducted in Indian protected areas, including cattle grazing in protected areas, unlawful trades in forest goods, and wildlife encroachment that negatively impacts community livelihoods.

In the Indian communities that around these protected regions, crop destruction is a prevalent issue. In most of these regions, crop damage is mostly caused by elephants, rhinoceros, wild boars, deer, monkeys, snow leopards, and parakeets, according to research. These creatures consume

a wide range of crops, including yams, potatoes, taro, and wheat as well as subterranean tubers like lentils and rice. Villagers employ a variety of strategies, including ditches, fences, and age-old techniques, to keep these wild animals away from their crops. No one strategy works for all of these animals due to variations in their sizes, food preferences, the kinds of crops they target, and the distances from farms. For instance, because of their size and eating habits, elephants cannot be deterred by the same techniques that is used to prevent deer. In a similar vein, since wild boars prey on subterranean tubers, strategies for managing them must diverge from those for deterring deer. Though a lot of study has been done on the issue of crop damage, not as much has been done to demonstrate how successful the methods for managing animal damage that are now in use are. Due to habitat loss, natural prey depletion, and deforestation, wild animals are forced to reside near to human settlements and eat domestic cattle. The creatures then started to intrude into human territory, endangering livestock, crops, and occasionally, humans and their farms.

This work aims at developing an animal intrusion detection system, that can precisely identify the intruding animals at a considerable distance, alert farmers well in advance as well as rise alarms to drive away the intruding animals.

II. RELATED WORKS

Various methodologies and advanced deep learning models have been investigated to enhance the accuracy and efficiency of the proposed work.

Authors in [1] have utilized YOLO object detection to identify wild animals near populated areas, achieving high accuracy for human and animal detection. Practical implementation on Raspberry Pi devices is emphasized. Authors in [2] cross-camera tracking is emphasized while using MobileNet and ResNet topologies for animal detection and re-identification. The paper obtained good accuracy findings, notably for tigers, jaguars, and elephants, with 80%, 89.47%, and 92.56% accuracy for animal detection Jeevitha et al. [3] introduced an IoT monitoring network system with wireless sensors for detecting animal intrusion. The system captures images and uses image processing to classify intruders, sending automatic alerts to landowners and forest officials via GSM. Narula et al. [4] proposed a sensor-based system for smart irrigation and animal detection in farming. The system aims to assist farmers in addressing issues like animal crop raiding, water conservation, and enhancing agricultural yield through intelligent irrigation.

Authors in [5] used PIR and ultrasonic sensors to detect animals in farm regions. The system achieves a 90% detection accuracy, even in challenging conditions, offering a cost-effective and simple installation. Authors in [6] identifies animal invasions using a network of cameras and real-time animal recognition algorithms. The system has a 90% accuracy rate. Chen et al. [7] has a 90% detection accuracy when using PIR and ultrasonic sensors to find animals in agricultural areas. Redmon et al. [8] explains how to evaluate the performance of object detection models using mean Average Percentage Error (mAP) and provides an overview of many deep learning algorithms used in object detection.

Authors in [9] introduced a system using Convolutional Neural Networks (CNNs) for identifying wild animals in agriculture farms. The system achieves a 95% accuracy rate in real-time settings. Sabeenian et al. [10] combines sensors, cameras, and AI-powered devices to detect and repel wild animals with high accuracy, providing effective solutions for preventing animal intrusions in protected areas. Authors in [11] presents a practical approach for protecting crop fields by studying animal behavior, detecting intruders, emitting irritating sounds to deter animals, and alerting authorized personnel with zero false alarm rates and accurate species identification. Authors in [12] introduces a Security System to the identification model. In [13-15], classification algorithms in various domains have been studied.

Rest of the paper is organized as follows: Section III discussed about the dataset used in the work. Section IV details the proposed methodology. Results are presented in Section V and finally conclusion and future work are provided in Section VI.

III.DATASET

The data for the proposed work was collected with the original goal of training an embedded device to perform real-time animal detection in nature reserves in South Africa.

Four animal classes commonly found in nature reserves in South Africa are represented in this data: buffalo, elephant, rhino and zebra.

This data contains at least 376 images for each animal class collected via Google's image search function and labelled for object detection. Each example in the data consists of a jpg image and a txt label file. The images have differing aspect ratios and contain at least one example of the specified animal class. Multiple instances of animals can exist in a single image. There could also be occurrences of the other classes in the same image, e.g. a zebra(3) in the file with an Bufflo(1) as shown in Fig. 1

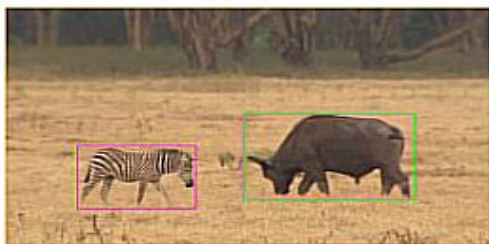


Fig 1. Sample image of Dataset [1]

IV.PROPOSED METHODOLOGY

The most important aspect of this study is choosing a model, and to address the issues of efficiency and complexity, YOLO v8, the most recent iteration of Ultralytics popular YOLO [16] (You Only Look Once) series of models, which are renowned for their speed and accuracy is chosen. YOLOv8 builds on the success of earlier YOLO versions, adding new features and improvements to further increase performance and flexibility. One of YOLO v8's key features is its anchor-free detection system, which eliminates the need for the model to pre-define an anchor set.

TABLE I. COMPARISON OF YOLO VERSIONS

Model	Parameters (M)	Latency A100 TensorRT FP16 (ms/img)	COCO MAP50-95 Val
YOLO v8	45	3	55
YOLO v7	35	2.5	50
YOLO v6-2.0	30	2	45
YOLO v5-7.0	25	1.5	40

Table I shows the relationship between the values of the parameters and the latency of the sensor. The values of the parameters are:

- Model: The name of the object detection model.
- Parameters (M): The number of parameters in the model.
- Latency A100 TensorRT FP16 (ms/img): The latency of the model when running on an A100 TensorRT FP16 device.
- COCO MAP50-95 Val: The mean average precision (mAP) of the model on the COCO validation dataset.

The table shows that the YOLOV8 model has the highest mAP, but also the highest latency. The YOLOV5-7.0 model has the lowest latency, but also the lowest mAP. There is a trade-off between latency and accuracy, and the best model for a particular application will depend on the specific requirements.

Bounding boxes for every object class, as mandated by earlier iterations of YOLO. Yolo v8 is now more effective and flexible to various sizes and shapes of objects. YOLO v8 additionally comprises several other enhancements, like:

- Novel design for a convolutional block that increases efficiency and accuracy.
- Data augmentation using mosaics augmenting method that can enhance the generalizability of the model.
- Assistance with a broader variety of object detection tasks involving segmentation, etc.

YOLO v8 comes in an array of sizes and speeds, so it is appropriate for a variety of uses, such as instantaneous

object identification on mobile devices to a highly precise picture partitioning of cloud servers.

V. WORK FLOW

A. Data Pre-Processing

Images are resized to a consistent dimensions, pixel values are normalized and data augmentation techniques are used to increase dataset size (e.g., random rotations, flips, and color adjustments). Images of all the classes are renamed according to their class which can be buffalo, elephant, rhino or zebra.

B. Creating Working Directories

The initial setup of establishing working directories and structuring image and label files for a multi-object detection project is done. Then, a hierarchical directory structure to store the data, segregating it into training, validation, and test sets is performed. These directories are constructed based on the current working directory. Next, a specific paths for images and labels within each of these subsets is defined ensuring a systematic organization that facilitates efficient data management. Then the working space is populated with images and labels corresponding to various animal classes. It iterates through a specified directory, likely containing subdirectories for each animal category. For label files, they are duplicated into the pertinent label directory and renamed with a prefix denoting the respective class.

C. Extracting And Visualising Test Images

A difficult test set is created to verify how good the proposed model can perform. A difficult test set is created to verify how good the proposed model can perform. Of all the images, those images in test set which have objects of different classes is taken. So all test images will have at least 2 objects of different classes. For example, a buffalo and a zebra in a single image is qualified to be part of the test data set. There can be multiple objects too. So 2 elephants and 1 buffalo in a single image is also perfectly valid. For the training and validation set, we will have images of only one object class. Any image in training and validation set can have one type of animal only, say rhino. Although there can be multiple rhinos in the same image.

D. Separating Training And Validation Dataset

This illustrates a critical phase in the development of a machine learning model for object detection. Specifically, it addresses the partitioning of a training dataset into distinct training and validation datasets. This division, achieved by moving 20% of the data to the validation set, is a fundamental practice in machine learning to assess model performance, prevent overfitting, and ensure the generalizability of the model. During this process, the code efficiently manages image and label files, relocating them to the respective training and validation directories. The subsequent verification of dataset sizes confirms that the split has been successfully executed. This meticulous dataset preparation is essential for training and evaluating machine learning models

E. Image Visualisation With Bounding Box

The primary purpose is to visualize bounding boxes on images. It accomplishes this by randomly selecting images

from a training dataset and their corresponding label files. The code reads the selected image, creates a duplicate, and overlays the bounding boxes, drawn in red, onto the duplicate. The result is a pair of images – the original and the annotated version – showcasing the objects within the images and their precise spatial positioning.

F. Model Training

YOLO models are trained to identify and draw bounding boxes around objects of interest. YOLO model is initialized using a configuration file named 'yolov8m.yaml'. The configuration file likely contains details about the model's architecture, hyperparameters, and other settings.

The model is trained with 100 epochs. which implies that the model undergoes 100 complete passes through the training dataset.

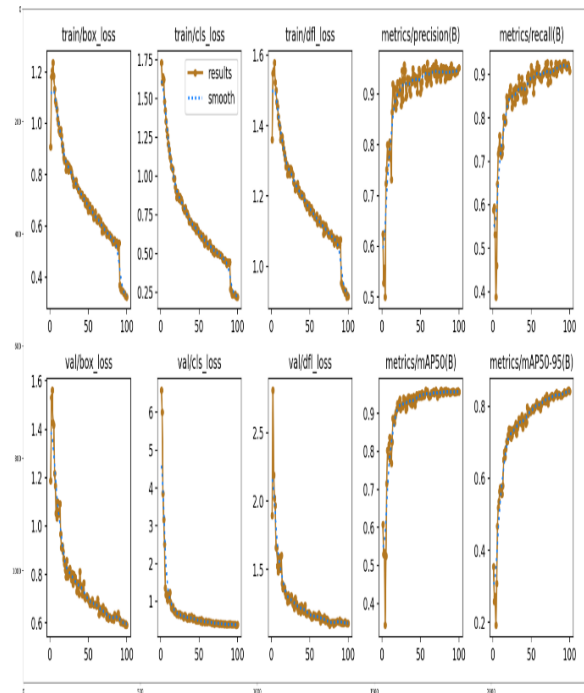


Fig 2. The training progressed with epochs

In the Fig.2, the train loss is initially much higher than the validation loss. This is because the model is randomly initialized at the beginning of training, and it takes some time for the model to learn to make accurate predictions.

The train loss decreases rapidly in the early stages of training. This is because the model is able to quickly learn some basic patterns in the training data.

The train loss decreases more slowly in the later stages of training. This is because the model is now learning more complex patterns in the training data.

The validation loss starts to decrease after a few epochs. This suggests that the model is starting to generalize to new data.

VI. EXPERIMENT AND RESULTS

The primary experiments in this study were conducted using the Ultralytics framework in conjunction with Python. Local experiments were executed on a GTX 1050 GPU.

TABLE II. THE MAP OF MODEL FOR ALL IMAGES ON TRAIN DATASET IS 0.994

Class	Instances	Box(P)	R	mAP50	mAP50-95
All	2073	0.996	0.992	0.994	0.964
Buffalo	435	0.997	1	0.995	0.972
Elephant	584	0.994	0.993	0.995	0.956
Rhino	440	0.998	0.995	0.995	0.977
zebra	614	0.995	0.98	0.993	0.94

During the training phase, which encompassed 100 epochs of Yolo v8, each individual training epoch consumed approximately 1 minute and 45 seconds. Consequently, the aggregate training time required to construct a single model amounted to roughly 1.5 hours. The training and validation accuracies of the model is shown in the Fig. 3,4 and Fig. 6,7.

The confusion matrix Fig.5 and Fig. 8 is also plotted to determine the performance of the model regarding False positives and False Negatives.

The Table II and Table III shows overall result of train and validation data. In Table II and Table III class 'all' in addition to the 4 classes. The field 'all' here aggregates the true positives, false positives and false negatives across all classes to calculate a single precision and recall value for the entire dataset.

Calculation of P and R are defined for 'all' below:

1. True Positives for Buffalo: TPb
2. False Positives for Buffalo: FPb
3. False Negatives for Buffalo: FNb
4. True Positives for Elephant: TPe
5. False Positives for Elephant: FPe
6. False Negatives for Elephant: FNe
7. True Positives for Rhino: TPr
8. False Positives for Rhino: FPr
9. False Negatives for Rhino: FNr
10. True Positives for Zebra: TPz
11. False Positives for Zebra: FPz
12. False Negatives for Zebra: FNz

Total True Positives

$$(TPt) = TPb + TPz + TPr + TPe \quad (1)$$

Total False Positives

$$(Fpt) = FPb + FPz + FPr + FPe \quad (2)$$

Total False Negatives

$$(Fnt) = FNb + FNz + FNr + FNe \quad (3)$$

Now,

$$Pall = \frac{TPt}{(TPt + Fpt)} \quad (4)$$

$$Rall = \frac{TPt}{(TPt + Fnt)} \quad (5)$$

In Table II and Table III: mAP50 : mean Average Precision at 50, mAP50-95: mean Average Precision between 50-95

TABLE III. THE MAP OF MODEL FOR ALL IMAGES ON VALIDATION DATASET IS 0.959

Class	Instances	Box(P)	R	mAP50	mAP50-95
All	540	0.95	0.915	0.959	0.845
Buffalo	97	0.948	0.945	0.96	0.909
Elephant	151	0.925	0.861	0.93	0.762
Rhino	114	0.975	0.965	0.98	0.93
zebra	178	0.952	0.89	0.94	0.78

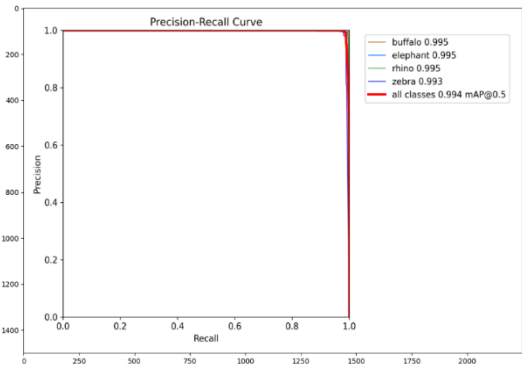


Fig 3. Precision – Recall curve of training data

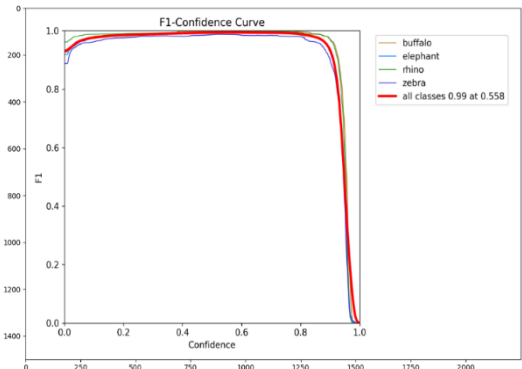


Fig 4. F1-Confidence curve of training data

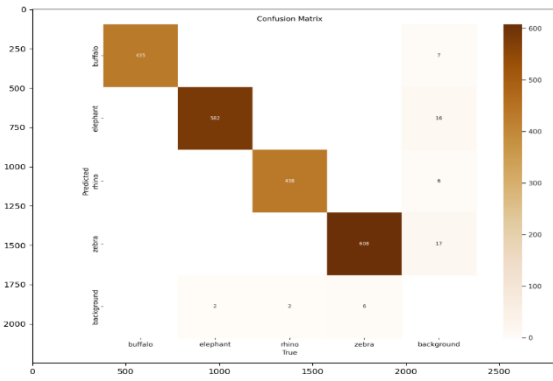


Fig 5. Confusion matrix of training data

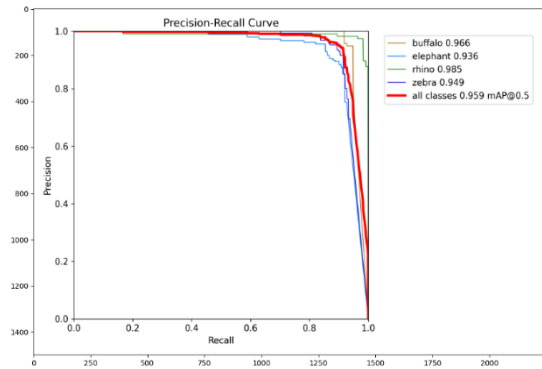


Fig 6 Precision – Recall curve of validation data

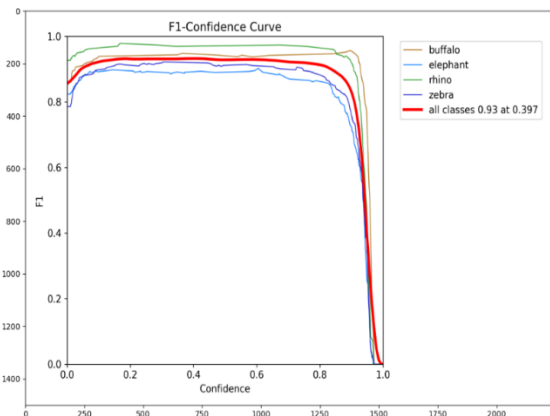


Fig 7 F1-Confidence curve of validation data

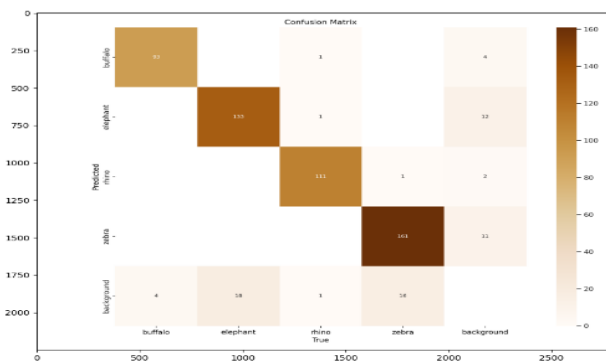


Fig 8. Confusion matrix of validation data

VII.CONCLUSION AND FUTURE WORKS

This work seamlessly uses YOLO v8 object detection for an Animal Intrusion Detection System, effectively safeguarding agricultural resources and managing human-wildlife conflicts. It can overcome real-time performance challenges, ensuring system reliability with a user-friendly interface. By preventing crop damage, it contributes to sustainable agriculture. The implementation demonstrates advanced technology's efficacy in solving practical challenges, fostering harmony between agriculture and the environment for future resource preservation. Enhancements like species identification, behavioral analysis, real-time tracking, and community engagement offer avenues for further research and application. Legal and ethical considerations are crucial for comprehensive solutions, promoting coexistence and mitigating conflicts.

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